

# Daniel Alvarado Peláez



**Degrees:** Master in Cybersecurity | Mechatronic Engineer

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## About me

I am an Ecuadorian **Mechatronics Engineer and Master in Cybersecurity**, passionate about technology and innovation. I have extensive experience in the design, development, and implementation of high-impact solutions, integrating **software development, cybersecurity, data analysis, and artificial intelligence** to solve complex challenges, with a focus on enhancing business productivity through ethical and efficient practices.

My profile combines a solid technical background with the ability to **lead technology teams and projects** throughout their entire lifecycle: from data ingestion to full-stack development, architecture, automation, and security audits. I am characterized by my analytical mindset, results-driven approach, and adaptability within **multidisciplinary and multilingual environments**, allowing me to successfully integrate into global and multicultural work dynamics.

## Experience

### General Specialist

Information Tecnology XOA S. A. / MAY 2023 - Present

- Full-stack development: design and implementation of secure web applications using modern technologies and coding best practices.
- Machine Learning model development and integration with company APIs and enterprise services, including production ML pipelines.
- Data analysis and processing: defining strategies for acquisition, cleaning, preprocessing, and handling large-scale data volumes with specialized tools.
- Information security: implementing protection measures, code audits, input validation, secure credential management, and service hardening.
- Security improvements: identifying vulnerabilities, assessing risks, designing mitigation solutions, and ensuring regulatory compliance.
- Technical support: field work for diagnostics, problem detection, and critical infrastructure issue resolution at client sites.
- Proactive monitoring: continuous supervision of services, servers, and production applications to ensure availability, performance, and security.
- Version updates and migrations: planning and implementing secure updates with minimal downtime and comprehensive validation.
- Internal tool development: creating company-specific services and utilities to optimize operational processes.

- Quality control: functional, security, and performance testing for embedded systems and web applications before release.

## **Research Assistant (Development Technician)**

**Center for Information Technologies (CTI) / MAR 2022 - APR 2022**

- Participation in a research project focused on predicting the energy consumption of Data Center infrastructure using Machine Learning techniques.
- Design and development of Jupyter notebooks for data acquisition, cleaning, preprocessing, and exploratory data analysis from monitoring systems.
- Implementation, training, and comparative evaluation of different predictive models, using performance metrics to determine the most efficient solution.
- Optimization and improvement of a data collection application, enhancing the quality, consistency, and availability of information used in analytical models.
- Collaboration in research activities that contributed to the scientific publication presented at IEEE Future Networks World Forum (FNWF 2022).

## **Web Developer**

**Edu4Lab / MAY 2021 - JUL 2021**

- Development of an educational web platform using Laravel under a scalable and maintainable system architecture.
- Implementation of user management, profile, and role modules, ensuring proper access control based on user responsibilities.
- Development of features for managing courses, books, online assessments, and academic staff.
- Design and integration of business logic and database for centralized management of educational content.
- Participation in functional testing and requirements validation to ensure proper platform operation.

## **Education**

### **Postgraduate, Master in Cybersecurity**

- Passed with honors; Cum laude

**Universidad Casa Grande / Guayaquil, Ecuador / JUL 2024 - JAN 2026**

### **Higher Education, Mechatronics Engineering**

**Escuela Superior Politécnica del Litoral (ESPOL) / Guayaquil, Ecuador / OCT 2017 - FEB 2023**

### **Primary - Secondary Education**

**Unidad Educativa FAE N°2 / Guayaquil, Ecuador / MAY 2005 - FEB 2017**

# Publications

## Comparison of Traditional ML Algorithms for Energy Consumption Prediction Models

2022 IEEE Future Networks World Forum (FNWF), Montreal, QC, Canada, 2022, pp. 232-237, R. Estrada, V. Asanza, D. Torres, I. Valeriano and D. Alvarado

doi: 10.1109/FNWF55208.2022.00048.



## Skills

### Programming languages

JAVA / Python / C/C++ / HTML / JavaScript / PHP

### Related

Node.js / Laravel / Jupyter Notebook / Dataiku / Astro / Vue.js

### Databases

MySQL / Postgresql / MongoDB / Redis / Meilisearch

### Related

Supabase / MongoDBCompass / MongoDB Atlas / Laragon / XAMPP / HeidiSQL / PGAdmin4 / DBeaver

### Other development tools

Git & GitHub / Docker / Cloudflare / Copilot / MATLAB & Simulink

### Cybersecurity

ISO 27001 / OWASP Top 10 / Personal Data Protection Law (Ecuador) / OSINT Tools / Shodan

### Related

Vulnerability analysis / Risk management / Security audits

## hardware tools

Arduino / Raspberry Pi / ESP32 - 8266 / PLC Siemens / PLC Logo

## Other tools

Paquete Office / Notion / Odoo / Claude / Copilot / Canva

## Technical skills

Development and programming / Analytics and data science / 3d design / Development of embedded systems

## Related

Autodesk Inventor / Autodesk AutoCAD / Automation Studio / CADe Simu / CCW (Connected Components Workbench) / FluidSIM / LabVIEW / TIA Portal / Proteus / FlexSim

## Personal skills

Teamwork / Commitment to tasks / Learning capacity / Orientation to results

## Languages

Spanish - Native / English - C1

## Projects

### Super Molar: Plaque Attack

Retro-styled run-and-gun platformer set inside a mouth, where a heroic tooth fights bacteria, cavities and bosses across five stages. All the pixel art is drawn in code on a 2D Canvas —there is not a single image file— and the sound effects and music are synthesised live with the Web Audio API. On top of that it runs a roguelike progression with random perks, four playable tooth classes and eight weapons that evolve up to level 5. It works entirely offline, with no telemetry and no network requests, and is localised in Spanish and English. Built with React 19, TypeScript, Tailwind v4 and Vite.



## RAPHAEL — The Grand Master

A knowledge entity presented as a construct of rings of light rather than as a chat interface. It is a single screen with no routing: the whole scene is drawn in WebGL2 from one render loop, so React never re-renders because of an animation. It answers through Gemini when an API key is present and, without one, from a local oracle that speaks in the same voice —the interface always tells one source from the other—. It includes a 107-parameter panel for tuning the scene live and accessibility limits enforced by tests (WCAG 2.3.1 and reduced motion in three registers). Built with React 19, TypeScript, Tailwind v4 and Vite.



## Thermal Monitoring Drone

CAD designs and Arduino circuits for a drone. CAD designs, circuits, and Raspberry Pi program for an embedded system that detects cable presence, captures thermal images, and detects temperature spikes.



## Real-time OPC Simulation

Real-time circuit simulation, connected via OPC to a LABVIEW interface, for monitoring and control.



## Proteus/Ubidots Pump Control

Design of a pump system in Proteus, controlled by an Atmega328p, and visualization in Ubidots of the system's status communicated by a program developed in Python.



## Knee Prosthesis Adaptation

3D design of a knee and editing of a total knee prosthesis, using Slice 3D and Blender software to adapt the existing knee prosthesis to the test knee.



## LABVIEW Pump Control

Design of an HMI for the control of a pump system, which has three pumps. Two pumps are primary and the third is backup in case one fails or is disabled for maintenance. Each pump has its start and stop pushbuttons, as well as its enable or disable selector. The system will have manual and automatic operation commanded by the signal sent by a selector.



## Dynamic Mechanism Simulation

A mechanism is designed where the shaft of a mill receives the necessary power and speed through pulley-chains. In turn, the shaft that transmits power to the pulley with chain receives the input power through a set of V-belt pulleys, as well as being supported by 2 bearings.



## Rice Stacker Design

Design and feasibility analysis of a rice stacker using renewable energy for implementation in a developing agricultural community.



## Automatic Pet Feeder

Automatic pet feeder project controlled by a mobile application that allows scheduling feeding times.



## Flexible Manufacturing Process

Redesigning the bottle manufacturing system of a company to reduce production times, increase system flexibility, and improve company profitability.



# **Certificates**

## **Despliegue de MySQL con Docker**

MAR 2023

## **Python para data scientist avanzado**

FEB 2023

## **Docker esencial**

FEB 2023

## **Learning Docker**

FEB 2023

## **OPENedX Escritura Académica**

ENE 2023

## **Examen CAMBRIDGE ENGLISH PLACEMENT TEST, C1**

DIC 2022

## **Cuarta revolución Industrial: Data science**

FEB 2022

## **Aprende Excel (Office365/Microsoft365)**

FEB 2022

## **Cómo eliminar las distracciones**

FEB 2022

## **Cómo conciliar las funciones múltiples del líder**

FEB 2022

## **Introducción a la programación en Python**

ABR 2021

## Fundamentos de programación en PLC

SEPT 2021

## Python para data science y big data esencial

SEPT 2021

## MATLAB Onramp

OCT 2020

## Certificate of Competency in English, ECCE B2

NOV 2017

## Certificado de graduación, CEN

MAY 2017



[Link to certificates](#)

## References

**Ph.D. Víctor Manuel Asanza - Professor of the FIEC faculty**

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